Software Test Plan (STP)

Change Management Software

Project	Change Management Software (CMS)
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Authors	Four-member Software Engineering project team
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Revision History

Version	Date	Author	Change summary	Approval
1.0	28 October 2026	Four-member Software Engineering project team	Initial project-specific draft	Pending

Approvals

Role	Name	Signature / Email	Date
Project team representative	To be filled by team	Pending	
Course instructor / TA	To be filled by instructor	Pending	

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1. Introduction

1.1 Purpose

This plan defines the scope, strategy, environment, responsibilities, and exit criteria for testing CMS. Tests verify that change requests are created and stored correctly, the status workflow allows only valid transitions, assignments and resolutions are recorded with full history, overdue requests are flagged accurately, and access controls are enforced.

1.2 Scope

Testing covers authentication/authorization, change request creation and editing, field validation, workflow transitions, approval and rejection, assignment and reassignment, comments and work notes, resolution, closure and reopening, status history, SLA/overdue detection, notifications, dashboard and reports, audit events, and key non-functional requirements. Email/chat delivery, source-control or CI/CD integration, and production-scale infrastructure are excluded unless implemented.

2. Test Items

- Authentication and role authorization
- Change request management module
- Workflow engine (status transitions)
- Assignment module
- Comments and activity timeline
- SLA / overdue detection
- Notification module
- Dashboard and reporting module
- Audit history
- Application routes, services, and persistence layer

3. Features to be Tested

Requirement / feature	Test focus
CMS-F-001 to CMS-F-002	Valid/invalid login, logout, role restrictions, direct protected-operation access.
CMS-F-003 to CMS-F-006	Create, view, edit, search, filter, and validation of CR fields; edits blocked on Closed CRs.
CMS-F-007 to CMS-F-008	Every allowed and disallowed workflow transition; rejection requires

	a reason.
CMS-F-009 / CMS-SR-005	Assignment to active engineers only; reassignment; engineers limited to their own CRs.
CMS-F-010 to CMS-F-012	Start work, work notes and comments, resolve with summary, close, and reopen with reason.
CMS-F-013	History rows for each status and assignment change, in order, with actor and timestamp.
CMS-F-014	Due-date calculation and overdue classification at boundary values.
CMS-F-015	Notifications created for assignment, status change and comment events.
CMS-F-016 to CMS-F-018	Dashboard counts, open-CR report, ageing and resolution-time report contents.
CMS-F-019 / CMS-SR-006	Audit event content and exclusion of secrets.
CMS-NF-001	Dashboard/search response time on the agreed test dataset.
CMS-NF-002 / CMS-NF-006	Persistence after restart and documented setup on a clean environment.
CMS-NF-003 / CMS-SR-001 to CMS-SR-004	Credential storage, authorization, validation, output escaping and safe error responses.
CMS-NF-007	Concurrent edits to the same CR are detected and not silently overwritten.

4. Features Not to be Tested

- Whether the change described in a CR was actually carried out in an external system; the system relies on user-entered data.
- Email, chat, source-control, CI/CD or other external integrations that are not part of the submitted build.
- Production infrastructure availability and load at enterprise scale.
- Third-party database and framework internals; test only the application's use of them.

5. Test Approach / Strategy

5.1 Test Levels

- Unit testing: validate service-level rules such as field validation, the workflow transition table, assignment eligibility, due-date calculation and overdue classification.
- Integration testing: validate service/repository/database interactions, transactional status and history writes, and optimistic locking.
- System testing: run end-to-end workflows (create, approve, assign, resolve, close, reopen) through the application interface or API.
- Acceptance testing: demonstrate the main user stories against the submitted SRS and verify the working product.

5.2 Test Types

- Functional testing of each implemented requirement.
- Negative and boundary-value testing for required fields, invalid transitions, permissions, due dates and empty data.
- State-transition testing covering every pair of statuses in the workflow table.
- Regression testing after service, UI, or schema changes.
- Security checks for authentication, authorization, record-level access, password storage, input validation, output escaping, and logs.
- Performance checks for dashboard/search using a documented dataset.
- Usability checks for clear navigation, labels, validation feedback, and status indicators.

5.3 Entry Criteria

- A runnable build is available and setup steps are documented.
- Test environment and test database are available.

- The team has agreed the implemented requirement set, the workflow transition table, and test data.
- Critical dependencies are installed and automated tests can be executed.

5.4 Exit Criteria

- All planned high-priority test cases have been executed.
- All critical defects are fixed or explicitly accepted by the team/instructor; no known critical workflow, history-integrity or authorization defect remains.
- Every test CR's history is consistent with its current status.
- Implemented requirements are linked to test cases in the RTM.
- Test results and deviations from SRS/SAD are documented.

6. Test Environment

Area	Plan
Hardware	Team development laptops/desktops; no special hardware required.
Software	The same Python runtime, virtual environment, application framework, and database specified by the project README.
Browser	Current version of a Chromium-based browser; add other browsers if required by the team.
Tools	Project unit-test framework (e.g. pytest), GitHub Actions CI, browser developer tools, and issue/PR tracking in GitHub.
Test data	Synthetic users for every role (including an inactive engineer); CRs in every status and priority; CRs due in the past, today, and in the future; a bulk set of 10,000 CRs for performance runs.
Isolation	Use a test database or resettable fixtures; do not run destructive tests against shared or real data.

7. Test Schedule

Activity	Target window	Output
Review SRS/SAD and prepare cases	Before / during Sprint 1	Test cases and RTM updates
Unit and integration tests for implemented stories	Sprint 1 (19–23 Oct 2026, per course brief)	Automated test results and defect list
End-to-end workflow regression, SLA and report tests	Sprint 2 (26–30 Oct 2026, per course brief)	Execution log and resolved defects
Final acceptance run and evidence capture	By end of Sprint 2	Test summary, screenshots/video evidence, final RTM

Dates above follow the course schedule supplied by the team. Adjust actual execution dates if work or submission dates change.

8. Test Deliverables

- This Software Test Plan.
- Test cases and test data/fixtures.
- Automated unit and integration test scripts.
- Test execution logs and CI results.
- Defect/issue records with reproduction steps.
- Requirements Traceability Matrix (RTM).
- Final Test Summary Report including pass/fail counts, known issues, and deviations.

9. Roles and Responsibilities

Role	Assigned person	Responsibilities
Test coordinator / QA	Team member 1 — fill in name	Maintain plan and RTM; coordinate test

		execution and report status.
Backend / workflow developer	Team member 2 — fill in name	Add unit tests for workflow, assignment and SLA services; fix service and data-integrity defects.
UI / integration developer	Team member 3 — fill in name	Verify UI workflows and integration behaviour; fix UI defects.
CI / documentation / cross-functional tester	Team member 4 — fill in name	Maintain CI checks, run regression tests, collect evidence, and update final documentation.

All four members should review one another's pull requests and run relevant tests before merging. These are suggested responsibilities; replace them with the team's actual assignments.

10. Risks and Mitigation

Risk	Mitigation
Late integration causes failures	Merge small PRs frequently and run CI on every PR.
Workflow tests miss a transition	Generate test cases from the transition table so every allowed and disallowed pair is covered.
Test data does not cover due-date boundaries	Include CRs due yesterday, today, tomorrow, and with no SLA set.
History drifts from current status	Use transactional service logic and a history-consistency test.
Tests depend on local configuration	Document environment setup and keep secrets out of the repository.
Requirements changed during implementation	Record each approved deviation and update SRS, RTM, and test cases.

11. Assumptions and Dependencies

- The project build and dependencies can be installed using the repository instructions.
- Test data is synthetic and can be reset between runs.
- All four team members have access to the shared repository and can review/merge changes according to the team's workflow.
- Any external integration not present in the build is treated as out of scope.

12. Suspension and Resumption Criteria

Suspend a test run if the application cannot start, the test database is unavailable, a blocker prevents more than 30% of planned cases from running, or a test risks corrupting shared data. Resume when the blocking defect is fixed, the environment is restored, and a smoke test passes.

13. Test Case Management and Traceability

Test case ID	Scenario	Expected result	Requirement
TC-AUTH-01	Sign in with valid credentials	User is authenticated and sees permitted dashboard.	CMS-F-001
TC-AUTH-02	Attempt protected action without permission	Request is rejected; no data changes.	CMS-F-002, CMS-SR-002
TC-CR-01	Create CR with valid required fields	CR is saved with unique ID and status Submitted, and is visible in search.	CMS-F-003
TC-CR-02	Submit CR with missing title	Validation message shown; no CR saved.	CMS-F-003, CMS-F-021
TC-VAL-01	Submit CR with a priority or component that does not exist	Request is rejected with validation feedback.	CMS-F-004
TC-VAL-02	Enter script text in CR description	Text is stored and displayed as plain text, not executed.	CMS-SR-004

TC-WF-01	Move CR through Submitted, Under Review, Approved, Assigned, In Progress, Resolved, Closed	Each step succeeds and status updates correctly.	CMS-F-007
TC-WF-02	Move a Submitted CR directly to Closed	Transition is rejected; status unchanged.	CMS-F-007
TC-WF-03	Reject a CR without entering a reason	Rejection is refused; status unchanged.	CMS-F-008
TC-ASSIGN-01	Assign an approved CR to an active engineer	Assignee set, status Assigned, engineer notified.	CMS-F-009, CMS-F-015
TC-ASSIGN-02	Assign a CR to an inactive user or a non-engineer	Assignment is rejected.	CMS-F-009
TC-ASSIGN-03	Engineer tries to resolve a CR assigned to someone else	Request is denied; no change.	CMS-SR-005
TC-RES-01	Mark CR Resolved without a resolution summary	Request is rejected; status unchanged.	CMS-F-011
TC-CLOSE-01	Requester closes a Resolved CR	Status Closed, closure date set; further edits blocked.	CMS-F-012, CMS-F-006
TC-REOPEN-01	Requester reopens a Resolved CR with a reason	Status Reopened; reason recorded; assignee notified.	CMS-F-012
TC-HIST-01	Complete full lifecycle including one reassignment	History lists every status and assignment change in order with actor and time.	CMS-F-013
TC-SLA-01	Open CR whose due date was yesterday	CR appears in overdue view.	CMS-F-014
TC-SLA-02	Resolved CR whose due date has passed	CR does not appear in overdue view.	CMS-F-014
TC-REP-01	Generate open-CR report by status and assignee	Report totals match stored CR data.	CMS-F-016, CMS-F-017
TC-REP-02	Generate resolution-time report for a date range	Only CRs closed in the range are included; average is correct.	CMS-F-018
TC-AUDIT-01	Approve and assign a CR	Audit records include actor/action/time and no secrets.	CMS-F-019, CMS-SR-006
TC-CONC-01	Two users edit the same CR using the same version	First update succeeds; second is rejected with a conflict message.	CMS-NF-007
TC-PERF-01	Measure 20+ dashboard/search requests on 10,000 CRs	At least 90% complete within 2 seconds.	CMS-NF-001
TC-SEC-01	Inspect stored credential representation and logs	No plaintext password or secret is stored/logged.	CMS-NF-003, CMS-SR-003
TC-RESTART-01	Restart application after committing data	Previously committed CRs, history and comments remain available.	CMS-NF-002

14. Test Metrics and Reporting

- Execution progress = executed test cases / planned test cases × 100%.
- Pass rate = passed test cases / executed test cases × 100%.
- Track failed, blocked, and not-run cases separately.
- Track open defects by severity, age, and affected requirement.
- Report requirement coverage as requirements with at least one test / implemented requirements.
- Report workflow coverage as transitions tested / transitions in the workflow table.
- Publish daily or per-PR status during the sprint and a final test summary before the demo.

15. Approvals

Role	Name	Signature / Date

Test coordinator / QA	To be filled by team	
Developer representative	To be filled by team	
Project representative	To be filled by team	
Instructor / TA	To be filled if required	